

Industrial IoT Safety & Energy-Saving System

Double-Verified Comprehensive Technical Specification & Wiring Audit

1. Double-Verification Audit & Architectural Sign-Off

This document details the exhaustive, double-verified hardware validation and multi-node mapping matrix for the smart factory safety infrastructure. To guarantee a zero-collision hardware environment, mitigate timing jitter on critical control signals, and bypass GPIO pin limitations inherent to individual microcontrollers, the system's operational workload is strictly partitioned across three independent processing nodes via isolated, asynchronous communication channels.

CRITICAL HARDWARE AUDIT VERIFICATION NOTE:

- 1. SPI1 Isolation: The MFRC522 RFID reader is isolated on Board #1. This keeps clock pulses up to 10MHz entirely separate from the matrix keypad's continuous row-grounding loops on Board #2.*
- 2. Independent Timer Registers: The gate servo (Board #1) and 12V exhaust fan (Board #2) both utilize pin PA8 (TIM1_CH1). Because they run on independent hardware units, they do not collide or disrupt individual register arrays.*
- 3. Dedicated Communication Bus: Local STM32 nodes stream telemetry concurrently to the ESP8266 Cloud Gateway over isolated UART links. This guarantees immediate, non-blocking delivery of Telegram alert payloads without polling delays.*

2. Final Verified Hardware Pin Connection Tables

The connection mappings below have been cross-audited against the native architecture configurations of the ESP8266 NodeMCU V3 and STM32F103C8T6 ARM Cortex-M3 microcontrollers. All pins are assigned to native peripheral channels to bypass software emulation bottlenecks.

Table 2.1: Central Cloud Gateway Node (1× ESP8266 NodeMCU V3)

ESP8266 Pin	Connected Device	Interface Type	Operational Mandate
TX (GPIO1)	STM32 Board #1 / #2 (RX)	UART Bus	Dispatches hardware clock sync signals and config overrides.
RX (GPIO3)	STM32 Board #1 / #2 (TX)	UART Bus	Receives asynchronous event strings and hazard alerts for Cloud output.
3.3V / GND	Regulated Power Rails	DC Bus Reference	Maintains common baseline reference plane to eliminate signal noise.

Table 2.2: Perimeter Access & Security Node (STM32 Board #1)

STM32 Pin	Target Component	Protocol Mode	Hardware Functional Role
PA4	MFRC522 RFID SDA	SPI1 NSS	Hardware chip select line for fast,

			synchronous RFID badge polling.
PA5	MFRC522 RFID SCK	SPI1 SCK	Outputs synchronous clock pulses up to 10MHz to drive data loops.
PA6	MFRC522 RFID MISO	SPI1 MISO	Accepts raw high-speed binary card token updates from badges.
PA7	MFRC522 RFID MOSI	SPI1 MOSI	Transmits configuration arrays and command bytes to reader.
PB0	MFRC522 RFID RST	Digital Out	Drives hardware reset pulse high/low during early initialization.
PA8	SG90 Gate Servo Motor	PWM (TIM1_CH1)	Generates a stable 50Hz PWM wave to cycle the physical gate arm.
PB6 / PB7	16x2 LCD Display + I2C	I2C1 (SCL/SDA)	Renders real-time operator names and project progress states.
PB1	Entrance PIR Sensor	Digital Input	Tracks entrance zones constantly for unverified walking patterns.
PB10	5V Active Buzzer Module	Digital Output	Emits quick validation chimes or 5s security alarm cycles.
PA0	Emergency Stop Button	Digital In (EXTI0)	High-priority external hardware interrupt for instantaneous cutoff.
PA9 / PA10	ESP8266 Gateway (RX/TX)	UART1 Bus	Streams security access and emergency alerts to cloud buffer.

Table 2.3: Machine Safety, Energy & Hazard Node (STM32 Board #2)

STM32 Pin	Target Component	Protocol Mode	Hardware Functional Role
PB3, 4, 5, 6	4x4 Keypad (Rows R1-R4)	Digital Output	Drives row lines LOW sequentially during active polling matrix loops.
PB7, 8, 9, 12	4x4 Keypad (Cols C1-C4)	Digital In Pull-up	Captures alphanumeric keypad entries via internal pull-up lines.
PB11	DHT11 Climate Sensor	Single-Wire	Decodes single-bus thermal and environmental humidity bounds.
PA0	MQ-4 Gas Sensor (AO)	Analog In (ADC1)	Converts methane/smoke concentrations into fine 12-bit values.
PA1	IR Flame Sensor (DO)	Digital Input	Monitors open-flame light waves for direct fire detection overrides.
PA2	Workshop PIR Sensor	Digital Input	Tracks continuous human movement to handle energy timeouts.

PA3, 4, 5, 6	4-Channel Relay (IN1-4)	Digital Output	Controls optoisolated relays to isolate machine main power grids.
PA8	12V DC Exhaust Fan	PWM (TIM1_CH1)	Modulates fan speed proportionally via external MOSFET drivers.
PB13, 14, 15	Status LEDs (R, G, Y)	Digital Output	Provides instant visual updates on localized system override states.
PA9 / PA10	ESP8266 Gateway (RX/TX)	UART1 Bus	Routes ambient sensor diagnostics and task tokens to gateway.

3. Event-Driven Execution & Automation Routines

3.1 Perimeter Access Checkpoint Workflows

- **Authorized Entry Sequence:** When an operator badge matches the firmware allowance array, the shared I2C LCD prints the personnel name and technical qualification. Simultaneously, the servo motor swings to 90 degrees, the active buzzer module fires a single short 150ms beep, and a secure event package is sent over UART to the ESP8266 to fire a Telegram notification.
- **Unauthorized Breached Blockade:** If an unlisted or malicious card UID is scraped, the entry gate servo is forced to lock at 0 degrees. The buzzer is engaged continuously for 5 seconds to notify guard staff, and an urgent security override text is routed to the manager's phone via the Telegram API loop.
- **Vagrant Roaming Interception:** The entrance PIR sensor continually scans for spatial motion. If human movement registers for 30 consecutive seconds without a successful RFID authorization token being validated, the microcontroller activates a loud 10-second defensive buzzer cadence and dispatches a 'Vagrant Roaming Near Gate' security alert to management.
- **Standard Egress Operations:** When exiting the building, employees tap their token a second time. The LCD clears and displays checkout verification text alongside the departing name, and an exit telemetry string logs the timestamp with the manager channel.

3.2 Multi-Factor Machine Operation & Task Handshakes

- **Secure Terminal Initialization:** To spin up an active industrial zone, the operator enters their secure alphanumeric sequence into the 4x4 matrix keypad. Once checked, the buzzer emits a validation beep, and the system queries internal memory to pull historical progress values. The LCD dynamically renders the worker name alongside an initial task status (randomly assigned as 25% or 50% based on the previous operator's shift state) before routing terminal telemetry to the cloud gateway.
- **Malicious Keypad Containment:** If an invalid security sequence is typed, the local LCD overwrites to state 'Unauthorized User'. The firmware trips the assigned relay pin to pull the optoisolated main machine line LOW, instantly cutting off all current flow to the heavy gear. A continuous buzzer tone is generated, and a security flag is routed to the supervisor.
- **Shift Sign-Off & Automation Metrics:** When wrapping up an operational shift, the worker taps the '#' key. This locks the terminal. The code automatically calculates operational progress metrics, assigns a random incremental bump to compile a final task accomplishment index (e.g., updating 25% to 75% or 50% to 100%), and fires a text report summarizing structural end states directly to the manager via Telegram.

3.3 Energy Conservation & Safety Override Interlocks

- **Occupancy Energy Automation:** Workshop PIR units monitor ongoing movement in the bay. If no human movement is recorded for 30 consecutive seconds, the MCU drives control lines high to disengage power relays. This isolates the non-essential lighting grid, floor ventilation fans, and idle machine terminal lines. Everything returns to a full power state the instant localized motion is captured again.

- **Proportional Climate Regulation:** The single-bus DHT11 updates workspace thermal readings constantly. The firmware evaluates updates against configured limits, mapping temperature shifts to a 12-bit hardware PWM array (TIM1_CH1). This smoothly scales the speed of the 12V DC exhaust fan, stabilizing room temperatures without human intervention or excessive power draw.
- **Multi-Hazard Main Breaker Tripping:** Atmospheric sensors continually screen for industrial safety anomalies using DHT11 thermal points, analog MQ-4 gas scanners, and digital optical IR flame units. If open fire spikes, a severe thermal runaway event, or high methane gas/smoke thresholds are broken, the MCU triggers global breaker relays to drop all line power across the factory floor, isolating downstream hazards and transmitting a red-alert hazard message to management channels.
- **Manual Hardware Interrupt E-Stop:** A high-visibility Emergency Stop push button is connected to hardware interrupt line EXTI0 (PA0). Tapping this switch instantly overrides active code paths, cuts power to all utilities, machinery, and lighting lines within microseconds, and streams a priority distress beacon via the cloud gateway.

4. System Interconnection & Electrical Topography

- **Power Regulation Network:** A main 12V 3A DC power adapter handles high-current loads, including the 4-channel relay coils and the 12V cooling fan. High-efficiency buck converters step the 12V rail down to 5V and 3.3V to power the logic chips, sensors, and microcontrollers without overheating.
- **Unified Ground Baseline:** To eliminate communications noise, signal floating, or false sensor triggers, all ground pins across the power supplies, microcontrollers, optoisolators, and sensors are tied into a single, unified reference ground plane.
- **Inductive Spike Protection:** High-speed 1N4007 flyback diodes are wired in parallel across all inductive components (including relay coils and DC fan terminals). This suppresses back-EMF voltage spikes, protecting the microcontrollers from sudden resets or hardware degradation during power shifts.

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